G2 Differentiating functions

Name:

Class:

Date:

Time: **344 minutes**

Marks: **302 marks**

Comments:

Q1.

A curve has equation $y = 2x^2 - 8x\sqrt{x} + 8x + 1$ for $x \geq 0$

(a) Prove that the curve has a maximum point at (1, 3)

Fully justify your answer.

(9)

(b) Find the coordinates of the other stationary point of the curve and state its nature.

(2)

(Total 11 marks)

Q2.

$$y = \frac{1}{x^2}$$

Find an expression for $\frac{dy}{dx}$

Circle your answer.

$\frac{dy}{dx} = \frac{0}{2x}$ $\qquad$ $\frac{dy}{dx} = x^{-2}$ $\qquad$ $\frac{dy}{dx} = -\frac{2}{x}$ $\qquad$ $\frac{dy}{dx} = -\frac{2}{x^3}$

(Total 1 mark)

Q3.

Find the coordinates of the stationary point of the curve with equation

$$(x + y - 2)^2 = e^y - 1$$

(Total 7 marks)

Q4.

A curve has equation $y = x^5 + 4x^3 + 7x + q$ where q is a positive constant.

Find the gradient of the curve at the point where $x = 0$

Circle your answer.

0 $\qquad$ 4 $\qquad$ 7 $\qquad$ q

(Total 1 mark)

Q5.

Chris claims that, "for any given value of x, the gradient of the curve $y = 2x^3 + 6x^2 - 12x + 3$ is always greater than the gradient of the curve $y = 1 + 60x - 6x^2$" .

Show that Chris is wrong by finding all the values of x for which his claim is **not** true.

(Total 7 marks)

Q6.

A curve has equation $y = 6x\sqrt{x} + \dfrac{32}{x}$ for $x > 0$

(a) Find $\dfrac{dy}{dx}$

(4)

(b) The point A lies on the curve and has x-coordinate 4

Find the coordinates of the point where the tangent to the curve at A crosses the x-axis.

(5)

(Total 9 marks)

Q7.

A particle, of mass 400 grams, is initially at rest at the point O.

The particle starts to move in a straight line so that its velocity, v m s $^{-1}$, at time t seconds is given by

$$v = 6t^2 - 12t^3 \text{ for } t > 0$$

(a) Find an expression, in terms of t, for the force acting on the particle.

(3)

(b) Find the time when the particle next passes through O.

(5)

(Total 8 marks)

Q8.

Find the coordinates, in terms of a, of the minimum point on the curve $y = x^2 - 5x + a$, where a is a constant.

Fully justify your answer.

(Total 3 marks)

Q9.

Prove that the function $f(x) = x^3 - 3x^2 + 15x - 1$ is an increasing function.

(Total 6 marks)

Q10.

A curve has equation $y = \frac{2}{\sqrt{x}}$

Find $\frac{dy}{dx}$

Circle your answer.

$\frac{\sqrt{x}}{3}$ $\qquad$ $\frac{1}{x\sqrt{x}}$ $\qquad$ $-\frac{1}{x\sqrt{x}}$ $\qquad$ $-\frac{1}{2x\sqrt{x}}$

(Total 1 mark)

Q11.

(a) Given that $u = 2^x$, write down an expression for $\frac{du}{dx}$

(1)

(b) Find the exact value of $\int_0^1 2^x\sqrt{3+2^x}\,dx$

Fully justify your answer.

(6)
(Total 7 marks)

Q12.

A curve has equation $y = \frac{2x+3}{4x^2+7}$

(a) (i) Find $\frac{dy}{dx}$

(2)

(ii) Hence show that y is increasing when $4x^2 + 12x - 7 < 0$

(4)

(b) Find the values of x for which y is increasing.

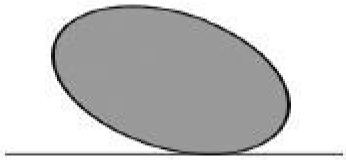
(2)
(Total 8 marks)

Q13.

A sculpture formed from a prism is fixed on a horizontal platform, as shown in the diagram.

The shape of the cross-section of the sculpture can be modelled by the equation $x^2 + 2xy + 2y^2 = 10$, where x and y are measured in metres.

The x and y axes are horizontal and vertical respectively.



Find the maximum vertical height above the platform of the sculpture.

(Total 8 marks)

Q14.

An open-topped fish tank is to be made for an aquarium.
It will have a square horizontal base, rectangular vertical sides and a volume of 60 m^3
The materials cost:

- £15 per m^2 for the base
- £8 per m^2 for the sides.

(a) Modelling the sides and base of the fish tank as laminae, use calculus to find the height of the tank for which the overall cost of the materials has its minimum value.

Fully justify your answer.

(8)

(b) (i) **In reality, the thickness of the base and sides of the tank is 2.5 cm**

Briefly explain how you would refine your modelling to take account of the thickness of the sides and base of the tank.

(1)

(ii) How would your refinement affect your answer to part **(a)**?

(1)

(Total 10 marks)

Q15.

A curve is defined by the parametric equations

$$x = t^3 + 2 , \qquad y = t^2 - 1$$

(a) Find the gradient of the curve at the point where $t = -2$

(4)

(b) Find a Cartesian equation of the curve.

(2)

(Total 6 marks)

Q16.

The equation $x^3 - 3x + 1 = 0$ has three real roots.

(a) Show that one of the roots lies between −2 and −1

(2)

(b) Taking $x_1 = -2$ as the first approximation to one of the roots, use the Newton-Raphson method to find x_2, the second approximation.

(3)

(c) Explain why the Newton-Raphson method fails in the case when the first approximation is $x_1 = -1$

(1)
(Total 6 marks)

Q17.

A curve has equation $y = 2x \cos 3x + (3x^2 - 4) \sin 3x$

(a) Find $\frac{\mathrm{d}y}{\mathrm{d}x}$, giving your answer in the form $(mx^2 + n) \cos 3x$, where m and n are integers.

(4)

(b) Show that the x-coordinates of the points of inflection of the curve satisfy the equation

$$\cot 3x = \frac{9x^2 - 10}{6x}$$

(4)
(Total 8 marks)

Q18.

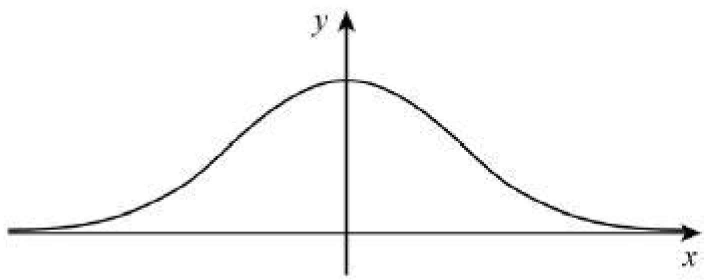
$\int_1^2 x^3 \ln(2x)\mathrm{d}x$ can be written in the form $p\ln 2 + q$, where p and q are rational numbers.

Find p and q.

(Total 5 marks)

Q19.

The diagram shows part of the graph of $y = e^{-x^2}$

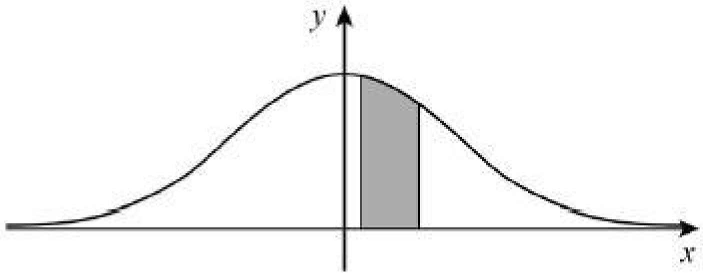


The graph is formed from two convex sections, where the gradient is increasing, and one concave section, where the gradient is decreasing.

(a) Find the values of x for which the graph is concave.

(4)

(b) The finite region bounded by the x-axis and the lines x = 0.1 and x = 0.5 is shaded.



Use the trapezium rule, with 4 strips, to find an estimate for $\int_{0.1}^{0.5} e^{-x^2}\,dx$

Give your estimate to four decimal places.

(3)

(c) Explain with reference to your answer in part **(a)**, why the answer you found in part **(b)** is an underestimate.

(2)

(d) By considering the area of a rectangle, and using your answer to part **(b)**, prove that the shaded area is 0.4 correct to 1 decimal place.

(3)
(Total 12 marks)

Q20.

The point $P(2, 8)$ lies on a curve, and the point M is the only stationary point of the curve.

The curve has equation $y = 6 + 2x - \frac{8}{x^2}$.

(a) Find $\frac{dy}{dx}$.

(3)

(b) Show that the normal to the curve at the point $P(2, 8)$ has equation $x + 4y = 34$.

(3)

(c) (i) Show that the stationary point M lies on the x-axis.

(3)

(ii) Hence **write down** the equation of the tangent to the curve at M.

(1)

(d) The tangent to the curve at M and the normal to the curve at P intersect at the point T. Find the coordinates of T.

(2)
(Total 12 marks)

Q21.

(a) Find $\frac{dy}{dx}$ when

$y = e^{3x} + \ln x$

(2)

(b) (i) Given that $u = \frac{\sin x}{1 + \cos x}$, show that $\frac{du}{dx} = \frac{1}{1 + \cos x}$.

(3)

(ii) Hence show that if $y = \ln\left(\frac{\sin x}{1 + \cos x}\right)$, then $\frac{dy}{dx}$ = cosec x.

(2)
(Total 7 marks)

Q22.

A curve has equation $y = x^5 - 2x^2 + 9$. The point P with coordinates (−1, 6) lies on the curve.

(a) Find the equation of the tangent to the curve at the point P, giving your answer in the form $y = mx + c$.

(5)

(b) The point Q with coordinates (2, k) lies on the curve.

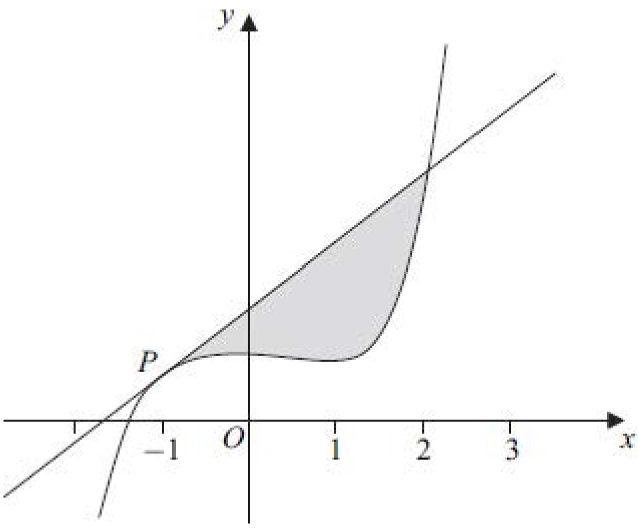
(i) Find the value of k.

(1)

(ii) Verify that Q also lies on the tangent to the curve at the point P.

(1)

(c) The curve and the tangent to the curve at P are sketched below.



(i) Find $\int_{-1}^{2} (x^5 - 2x^2 + 9)\,dx$.

(5)

(ii) Hence find the area of the shaded region bounded by the curve and the tangent to the curve at P.

(3)

(Total 15 marks)

Q23.

A curve has the equation

$$y = \frac{12 + x^2\sqrt{x}}{x}, \quad x > 0$$

(a) Express $\frac{12 + x^2\sqrt{x}}{x}$ in the form $12x^p + x^q$.

(3)

(b) (i) Hence find $\frac{dy}{dx}$.

(2)

(ii) Find an equation of the normal to the curve at the point on the curve where $x = 4$.

(4)

(iii) The curve has a stationary point P. Show that the x-coordinate of P can be written in the form 2^k, where k is a rational number.

(3)

(Total 12 marks)

Q24.

(a) Given that $y = x^4 \tan 2x$, find $\frac{dy}{dx}$.

(3)

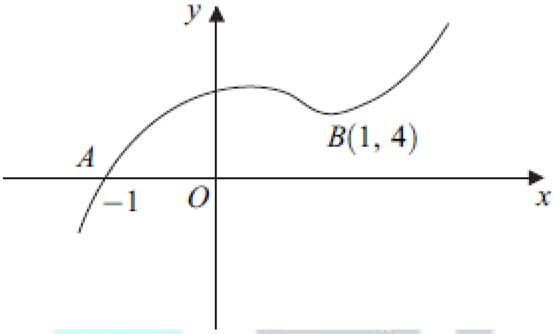
(b) Find the gradient of the curve with equation $y = \frac{x^2}{x-1}$ at the point where $x = 3$.

(3)

(Total 6 marks)

Q25.

The curve with equation $y = x^5 - 3x^2 + x + 5$ is sketched below. The point O is at the origin and the curve passes through the points $A(-1, 0)$ and $B(1, 4)$.



(a) Given that $y = x^5 - 3x^2 + x + 5$, find:

(i) $\frac{dy}{dx}$;

(3)

(ii) $\frac{d^2y}{dx^2}$.

(1)

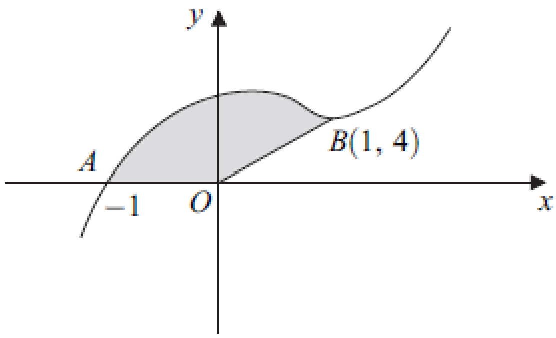
(b) Find an equation of the tangent to the curve at the point $A(-1, 0)$.

(2)

(c) Verify that the point B, where $x = 1$, is a minimum point of the curve.

(3)

(d) The curve with equation $y = x^5 - 3x^2 + x + 5$ is sketched below. The point O is at the origin and the curve passes through the points $A(-1, 0)$ and $B(1, 4)$.



(i) Find $\int_{-1}^{1} (x^5 - 3x^2 + x + 5)\,dx$.

(5)

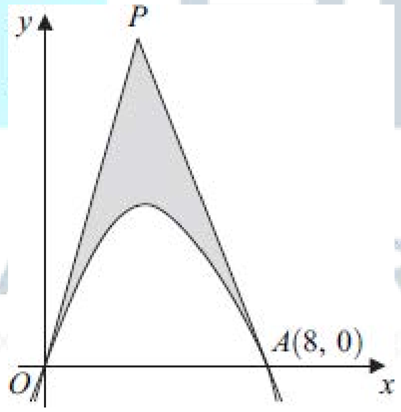
(ii) Hence find the area of the shaded region bounded by the curve between A and B and the line segments AO and OB.

(2)

(Total 16 marks)

Q26.

The diagram shows part of a curve crossing the x-axis at the origin O and at the point $A(8, 0)$. Tangents to the curve at O and A meet at the point P, as shown in the diagram.



The curve has equation

$$y = 12x - 3x^{\frac{5}{3}}$$

(a) Find $\frac{dy}{dx}$.

(2)

(b) (i) Find the value of $\frac{dy}{dx}$ at the point O and hence write down an equation of the tangent at O.

(2)

(ii) Show that the equation of the tangent at $A(8, 0)$ is $y + 8x = 64$.

(3)

(c) Find $\int \left(12x - 3x^{\frac{5}{3}}\right) \mathrm{d}x$.

(3)

(d) Calculate the area of the shaded region bounded by the curve from O to A and the tangents OP and AP.

(7)
(Total 17 marks)

Q27.

(a) Given that $y = 4x^3 - 6x + 1$, find $\frac{\mathrm{d}y}{\mathrm{d}x}$.

(1)

(b) Hence find $\int_2^3 \frac{2x^2 - 1}{4x^3 - 6x + 1} \mathrm{d}x$, giving your answer in the form $p \ln q$,

where p and q are rational numbers.

(5)
(Total 6 marks)

Q28.

(a) Given that $x = \frac{1}{\sin\theta}$, use the quotient rule to show that $\frac{\mathrm{d}x}{\mathrm{d}\theta} = -\operatorname{cosec}\theta\cot\theta$.

(3)

(b) Use the substitution $x = \operatorname{cosec}\theta$ to find $\int_{\sqrt{2}}^{2} \frac{1}{x^2\sqrt{x^2 - 1}} \mathrm{d}x$, giving your answer to three significant figures.

(9)
(Total 12 marks)

Q29.

At the point (x, y), where $x > 0$, the gradient of a curve is given by

$$\frac{\mathrm{d}y}{\mathrm{d}x} = 3x^2 - \frac{4}{x^2} - 11$$

The point $P(2, 1)$ lies on the curve.

(a) (i) Verify that $\frac{\mathrm{d}y}{\mathrm{d}x} = 0$ when $x = 2$.

(1)

(ii) Find the value of $\frac{d^2y}{dx^2}$ when $x = 2$.

(4)

(iii) Hence state whether P is a maximum point or a minimum point, giving a reason for your answer.

(1)

(b) Find the equation of the curve.

(4)
(Total 10 marks)

Q30.

A curve has equation $y = x^3 \ln x$.

(a) Find $\frac{dy}{dx}$.

(2)

(b) (i) Find an equation of the tangent to the curve $y = x^3 \ln x$ at the point on the curve where $x = \mathrm{e}$.

(3)

(ii) This tangent intersects the x-axis at the point A. Find the exact value of the x-coordinate of the point A.

(2)
(Total 7 marks)

Q31.

The curve with equation $y = 13 + 18x + 3x^2 - 4x^3$ passes through the point P where $x = -1$.

(a) Find $\frac{dy}{dx}$

(3)

(b) Show that the point P is a stationary point of the curve and find the other value of x where the curve has a stationary point

(3)

(c) (i) Find the value of $\frac{d^2y}{dx^2}$ at the point P.

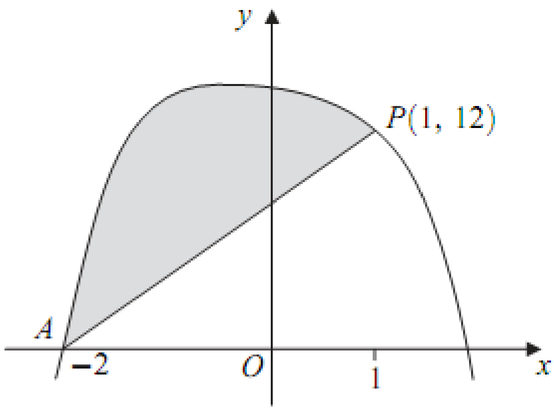
(3)

(ii) Hence, or otherwise, determine whether P is a maximum point or a minimum point.

(1)
(Total 10 marks)

Q32.

The curve sketched below passes through the point A (−2, 0).



The curve has equation $y = 14 - x - x^4$ and the point P (1, 12) lies on the curve.

(a) (i) Find the gradient of the curve at the point P.

(3)

(ii) Hence find the equation of the tangent to the curve at the point P, giving your answer in the form $y = mx + c$.

(2)

(b) (i) Find $\int_{-2}^{1} (14 - x - x^4)\,dx$.

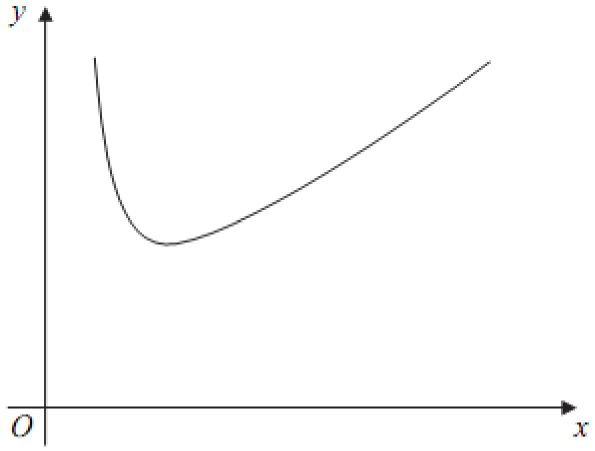
(5)

(ii) Hence find the area of the shaded region bounded by the curve $y = 14 - x - x^4$ and the line AP.

(2)
(Total 12 marks)

Q33.

A curve C is defined for $x > 0$ by the equation $y = x + 3 + \dfrac{8}{x^4}$ and is sketched below.



(a) Given that $y = x + 3 + \frac{8}{x^4}$, find $\frac{dy}{dx}$.

(3)

(b) Find an equation of the tangent at the point on the curve C where $x = 1$.

(3)

(c) The curve C has a minimum point M. Find the coordinates of M.

(4)

(d) (i) Find $\int \left(x + 3 + \frac{8}{x^4} \right) dx$.

(3)

(ii) Hence find the area of the region bounded by the curve C, the x-axis and the lines
$x = 1$ and $x = 2$.

(2)

(e) The curve C is translated by $\begin{bmatrix} 0 \\ k \end{bmatrix}$ to give the curve $y = f(x)$. Given that the x-axis is a tangent to the curve $y = f(x)$, state the value of the constant k.

(1)
(Total 16 marks)

Q34.

(a) Find $\frac{dy}{dx}$ when $y = (x^3 - 1)^6$.

(2)

(b) A curve has equation $y = x \ln x$.

(i) Find $\frac{dy}{dx}$.

(2)

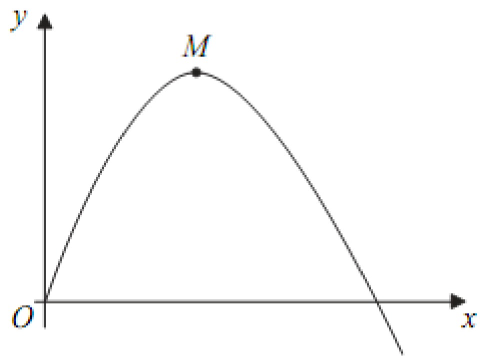
(ii) Find an equation of the tangent to the curve $y = x \ln x$ at the point on the curve where $x = e$.

(3)

(Total 7 marks)

Q35.

The diagram shows part of a curve with a maximum point M.



The curve is defined for $x \geq 0$ by the equation

$$y = 6x - 2x^{\frac{3}{2}}$$

(a) Find $\frac{dy}{dx}$.

(3)

(b) (i) Hence find the coordinates of the maximum point M.

(3)

(ii) Write down the equation of the normal to the curve at M.

(1)

(c) The point $P\left(\frac{9}{4}, \frac{27}{4}\right)$ lies on the curve.

(i) Find an equation of the normal to the curve at the point P, giving your answer in the form $ax + by = c$, where a, b and c are positive integers.

(4)

(ii) The normals to the curve at the points M and P intersect at the point R. Find the coordinates of R.

(2)
(Total 13 marks)